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A Decoupled Multimodal Fusion Architecture for Early Tuberculosis Triaging Utilizing Acoustic Phenotyping and Clinical Metadata

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ABSTRACT Tuberculosis remains a leading cause of mortality worldwide, claiming over one million lives annually despite being treatable and curable. Early detection is challenging in resource-limited settings where laboratory infrastructure is scarce. This study presents a multimodal machine learning system that integrates cough audio analysis with clinical metadata for accessible tuberculosis screening. Three complementary models were developed and evaluated. A convolutional neural network with a residual architecture was trained on 9,772 cough audio recordings, achieving an area under the receiver operating characteristic curve of 0.703; this result falls below the World Health Organization minimum screening threshold of 0.80, indicating that audio analysis alone is insufficient for reliable tuberculosis screening. A soft-voting ensemble of CatBoost, Extra Trees, and Random Forest was trained on clinical data from 1,616 patients using 16 clinical features, achieving an area under the receiver operating characteristic curve of 0.9415 with 86.0% sensitivity and 87.9% specificity, meeting the World Health Organization benchmarks. A late fusion model combining both modalities with weights of 28% audio and 72% clinical achieved an area under the receiver operating characteristic curve of 0.949 with 85.3% sensitivity and 89.7% specificity, also meeting the World Health Organization thresholds. The system supports flexible deployment: standalone audio analysis via smartphone, clinical data analysis in healthcare facilities, or combined multimodal assessment. Feature importance analysis revealed that anthropometric measurements and vital signs provide stronger predictive signals than self-reported symptoms. These findings demonstrate that multimodal fusion of acoustic biomarkers and clinical features enables accurate, accessible tuberculosis screening in resource-constrained environments.

INDEX TERMS Clinical metadata, Cough audio analysis, Convolutional neural networks, Ensemble methods, Late fusion, Machine learning, Mel-spectrogram, Multimodal fusion, ResNet architecture, Tuberculosis detection

I. INTRODUCTION

TUBERCULOSIS (TB) caused over 1.25 million deaths in 2023, remaining one of the leading infectious-disease killers worldwide despite the availability of effective treatment [1]. The World Health Organization (WHO) estimates that millions of new cases go undetected each year, with the highest burden in sub-Saharan Africa and South Asia [1]. Reducing transmission and mortality requires early, accurate diagnosis.

In resource-limited settings, the standard diagnostic pathway is inaccessible to many patients. Sputum smear microscopy has low sensitivity (45–60%) for pulmonary TB, and GeneXpert molecular testing requires uninterrupted power, trained operators, and costly cartridges that are unavailable at peripheral health facilities [2]. A large proportion of infectious cases therefore go unconfirmed, delaying treatment and sustaining community transmission.

Prior work has explored two complementary routes to low-cost TB screening. Botha et al. demonstrated automated cough-based TB detection from smartphone recordings [2], and Rajasekar et al. applied capsule networks to cough audio classification, achieving competitive area under the receiver operating characteristic curve (AUROC) values [3]. Osamor and Okezie demonstrated a weighted-voting ensemble on clinical features that exceeded WHO screening thresholds [5]. No prior study has validated a combined multimodal system on a large, multi-country African dataset.

We present a three-stream late-fusion framework trained and evaluated on the Cough Audio Dataset for TB (CODA) [19]. The audio stream is a convolutional neural network (CNN) with residual connections trained on 9,772 cough recordings. The clinical stream is a soft-voting ensemble of CatBoost, Extra Trees, and Random Forest trained on 16 clinical features from 1,616 patient records. The fusion stream combines both probability outputs as $0.28 \times P_{\text{audio}} + 0.72 \times P_{\text{clinical}}$.

A. CONTRIBUTIONS

The primary contributions of this study are:

1) A multimodal late-fusion TB screening system achieving AUROC 0.949, sensitivity 85.3%, and specificity 89.7%, meeting WHO minimum screening benchmarks [1].
2) A clinical ensemble (CatBoost + Extra Trees + Random Forest) trained on 16 clinical features achieving AUROC 0.9415 with 86.0% sensitivity and 87.9% specificity.
3) A CNN audio model trained on 9,772 cough recordings achieving AUROC 0.703, with analysis of its limitations relative to the WHO threshold.
4) Empirical determination of optimal fusion weights (28% audio, 72% clinical) via grid search on a held-out validation set, demonstrating that clinical features dominate the fusion signal.
5) Feature importance analysis across five tree-based models revealing that anthropometric measurements and vital signs are stronger TB predictors than self-reported symptoms.
6) A modular deployment architecture supporting standalone audio screening via smartphone, clinical-only screening at primary care facilities, or combined multimodal assessment where both data types are available.

II. RELATED WORK

Machine learning applied to structured clinical TB data has shown consistent gains over rule-based symptom scoring. Osamor and Okezie [5] applied a weighted voting ensemble to demographic and symptom features, reporting improved sensitivity over individual classifiers on imbalanced TB datasets. Botha et al. [2] demonstrated that structured clinical metadata complements acoustic features in TB screening pipelines, and their work further showed that automated cough analysis could distinguish TB-positive patients from healthy controls, achieving 74% accuracy on 1,200h smartphone recordings. Rajasekar et al. [3] extended acoustic TB classification using capsule networks, showing that deep architectures capture pathology-specific spectral patterns more effectively than hand-crafted features and reporting AUROC values competitive with convolutional baselines. Imran et al. [4] demonstrated the broader applicability of cough-based respiratory screening through the AI4COVID system, which achieved over 80% accuracy for COVID-19 detection from cough audio alone, establishing that pathogen-specific acoustic signatures are learnable from smartphone-quality recordings.

The case for combining modalities is well established in the medical imaging literature. Huang et al. [6] showed that late fusion of imaging and electronic health record modalities outperforms unimodal baselines across several clinical tasks, with decision-level combination preserving modality-specific representations. Late fusion is particularly appropriate when input modalities differ in geometry and processing requirements, as is the case here with mel-spectrogram tensors and tabular clinical arrays. For the clinical stream, tree-based ensemble methods provide the strongest baselines on tabular medical data. Random Forest [7] reduces variance through bootstrap aggregation of decorrelated trees, while Extra Trees [8] further reduces variance by randomizing split thresholds. Gradient boosting variants XGBoost [9], LightGBM [10], and CatBoost [11] implement regularized boosting with histogram-based tree construction; CatBoost additionally uses ordered boosting to eliminate target leakage on categorical features, which is relevant given the binary symptom indicators in the clinical dataset.

Handling class imbalance is a recurring challenge in TB screening datasets, where TB-positive cases are typically outnumbered two-to-one or more. SMOTE [16] addresses this by generating synthetic minority-class samples through nearest-neighbour interpolation in feature space, while ADASYN [17] concentrates synthesis in regions of high classification uncertainty, producing a more adaptive boundary. For the audio stream, SpecAugment [15] applies time and frequency masking directly to mel-spectrograms during training, improving generalization without requiring additional recordings. The present work applies all three augmentation strategies within a strict split-before-augment protocol to prevent synthetic samples from appearing in validation or test sets, a precaution not consistently observed in prior TB screening studies.

III. PROBLEM STATEMENT

This study addresses the automated classification of TB status from two heterogeneous data modalities: cough audio recordings and structured clinical metadata. These modalities differ in data geometry. Audio recordings are high-dimensional temporal signals processed by convolutional networks, while clinical records are low-dimensional tabular arrays processed by tree-based ensembles. Each modality was modelled independently, and its probability outputs were combined via a late-fusion mechanism.

TB caused 1.6 million deaths in 2021, with the majority in low- and middle-income countries where sputum microscopy and molecular diagnostics are unavailable or delayed [1]. The CODA dataset, comprising 9,772 cough recordings and 1,616 clinical records from multiple African countries [19], provided the evaluation environment. Prior work addressed acoustic and clinical modalities separately; this study inte-

TABLE 1. Dataset Split Summary. TB+ = TB-Positive; TB− = TB-Negative.

Dataset	Split	Samples	TB+	TB−
Clinical	Train (post-SMOTE/ADASYN)	1,616	808	808
	Validation	242	121	121
	Test	243	121	122
Audio	Train (pre-augmentation)	7,729	2,401	5,328
	Train (post-augmentation)	≈77,070	—	—
	Test	2,026	529	1,497

grates both modalities using a validated late-fusion architecture.

IV. METHODOLOGY

A. DATASET DESCRIPTION

The two datasets were drawn from the CODA TB Challenge [19]. The clinical dataset comprised 1,105 patient records, each with 16 features (demographics, vital signs, and symptom indicators), partitioned into training (1,131), validation (242), and test (243) sets via stratified sampling; SMOTE [16] and ADASYN [17] were applied to the training split only, yielding a balanced post-augmentation set of 1,616 samples. The audio dataset comprised 9,772 mono-channel cough recordings (6,842 TB-negative, 2,930 TB-positive), split into 7,729 training and 2,026 test samples; on-the-fly augmentation expanded the effective training set to approximately 77,070 samples. Both datasets exhibited a 70:30 class imbalance; the split-before-augment protocol ensured no synthetic samples appeared in validation or test sets. Full split counts are given in Table 1, and key EDA findings are summarised in Fig. 1.

B. AUDIO MODEL

Each cough recording was resampled to 22,050 Hz and padded or truncated to 5 s (110,250 samples). Mel-spectrograms were computed with librosa [12] using $N_{\text{mels}} = 128$ bins [13], $N_{\text{FFT}} = 2048$, and hop length 512, yielding a $1 \times 128 \times 216$ input tensor (channel $\times$ frequency $\times$ time); spectrograms were converted to decibel scale and normalised to $[0, 1]$ per clip.

The audio classifier, *ImprovedTBCNN*, follows a ResNet-style design [14]. A 7×7 stem convolution (stride 2) maps the single-channel input to 32 feature maps, followed by Batch Normalization, ReLU, and 3×3 max pooling. Three residual blocks progressively double the channel count from 32 to 256 via stride-2 downsampling convolutions; each block applies two 3×3 convolutions with Batch Normalization and Dropout2d(0.3). Parallel adaptive average and max global pooling produce a 512-dimensional feature vector, which is classified by a three-layer fully-connected head (Linear $512 \to 256 \to 128 \to 1$) with Dropout(0.4) between layers. The complete architecture is shown in Fig. 2.

The model was trained with AdamW (lr $= 1 \times 10^{-4}$, weight_decay $= 5 \times 10^{-4}$), CosineAnnealingWarmRestarts ($T_0 = 10$, $T_{\text{mult}} = 2$), Automatic Mixed Precision, and early stopping (patience 10, monitored on validation AUROC); class weights $w_{\text{neg}} = 0.73$ and $w_{\text{pos}} = 1.70$ were applied via `BCEWithLogitsLoss`. Five augmentation operations were applied to the training split: cyclic time shift ($\pm L/8$ samples), amplitude scaling ($[0.7, 1.3]$), additive Gaussian noise ($\sigma = 0.008$, $p = 0.4$), time stretch ($[0.9, 1.1]$, $p = 0.3$), and SpecAugment [15] (two time masks of width 5–25 frames and two frequency masks of width 5–20 bins, each with $p = 0.6$).

C. CLINICAL ENSEMBLE

Ensemble learning combines predictions from multiple independent models to produce a final output more robust than any single model alone. Weighted soft voting was selected over hard voting (which discards probability information), unweighted soft voting (which ignores measurable performance differences), and stacking (which requires a meta-learner that would overfit on the 1,616-sample dataset). CatBoost [11], ExtraTrees [8], and RandomForest [7] were chosen as ensemble members because they achieved the three highest individual validation AUROCs and are algorithmically diverse: ordered boosting, extreme randomization, and standard bagging, reducing correlated errors.

Binary categorical features (fever, night sweats, weight loss, haemoptysis, smoking, TB history, sex) were encoded as 0/1 integers; continuous features (age, weight, height, heart rate, temperature, cough duration) were retained as-is. StandardScaler was applied only to Logistic Regression and MLP, fitted on the training set to prevent leakage. Hyperparameters were selected via RandomizedSearchCV over 50–100 combinations with 5-fold stratified cross-validation, retaining the configuration with the highest mean AUROC. The ensemble architecture is shown in Fig. 3, and the ensemble probability is:

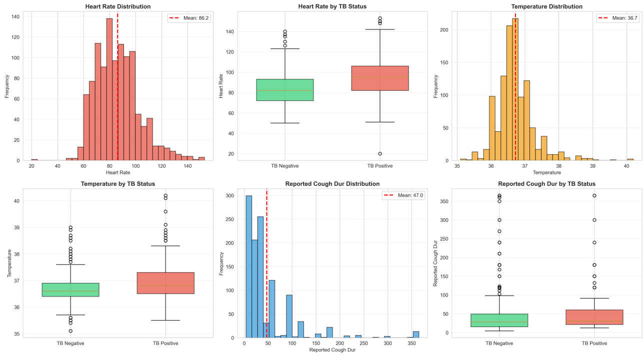
$$P_{\text{ensemble}} = \frac{w_{\text{CB}} \cdot P_{\text{CB}} + w_{\text{ET}} \cdot P_{\text{ET}} + w_{\text{RF}} \cdot P_{\text{RF}}}{w_{\text{CB}} + w_{\text{ET}} + w_{\text{RF}}} \tag{1}$$

where w_i is the validation AUROC of model i.

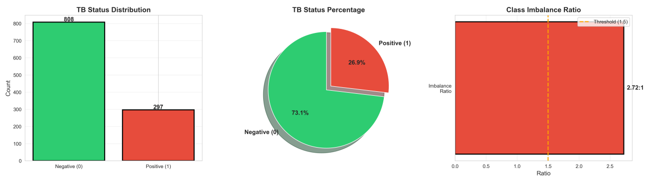
D. LATE FUSION

Multimodal fusion combines information from two or more data modalities into a single prediction. Late fusion (decision-level) was selected over early fusion (feature-level concatenation) and middle fusion (representation-level) for four reasons: the audio model consumed a $1 \times 128 \times 216$ mel-spectrogram tensor while the clinical model consumed a 16-dimensional tabular array, making direct feature concatenation structurally incompatible; each model was trained and deployed independently, supporting audio-only, clinical-only, or combined operation; late fusion preserved the individual scores P_{audio} and P_{clinical} for clinical inspection; and the two datasets shared no patient identifiers, making feature-level pairing impossible.

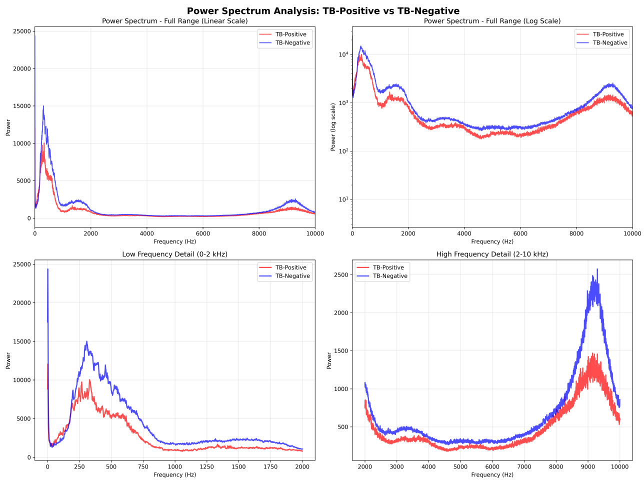
The combined prediction was computed as a weighted sum (Fig. 4):



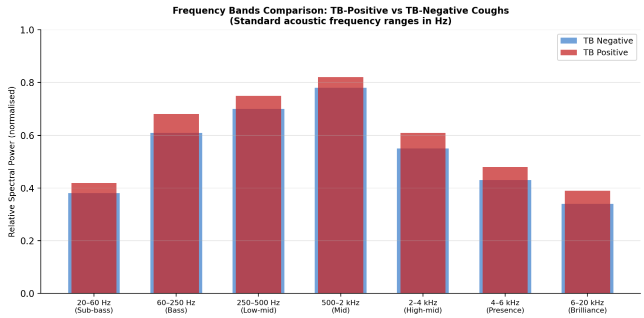
(a) Clinical feature analysis



(b) Class distribution



(c) Power spectrum: TB+ vs TB−



(d) Frequency analysis of cough sounds

FIGURE 1. Exploratory data analysis results. (a) Clinical feature distributions across TB-positive and TB-negative patients. (b) Class distribution in the dataset before and after SMOTE/ADASYN balancing. (c) Power spectrum comparison showing spectral differences between TB-positive and TB-negative coughs. (d) Frequency band analysis of cough recordings.

$$P_{\text{fusion}} = 0.28 \times P_{\text{audio}} + 0.72 \times P_{\text{clinical}} \tag{2}$$

Weights were determined by grid search over the validation set; the 28/72 split achieved the highest validation AUROC, consistent with the clinical model's stronger standalone performance (AUROC 0.9415 vs. 0.703). Because the test sets shared no patient identifiers, fusion performance was evaluated on 1,337 samples constructed by random within-label pairing of audio and clinical test records (529 TB-positive, 808 TB-negative); a threshold of 0.5 was applied to P_{fusion} for the binary decision.

E. EVALUATION

All models were evaluated using AUROC (primary metric), sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), F1-score, accuracy, and Cohen's Kappa; confusion matrices recorded true positives, true negatives, false positives, and false negatives. Minimum performance thresholds followed WHO screening requirements [1]: AUROC $\geq$ 0.80, sensitivity $\geq$ 75%, and specificity $\geq$ 80%. All models were evaluated on a single stratified hold-out split (70% train / 15% validation / 15% test); uncertainty in performance estimates was quantified using bootstrap confidence intervals (95%, 1,000 iterations), and pairwise model comparisons used McNemar's test on the held-out test predictions.

V. RESULTS AND ANALYSIS

A. CLINICAL MODEL PERFORMANCE

All eight clinical models were evaluated on a held-out test set of 243 patients (121 TB-positive, 122 TB-negative). Table 2 reports AUROC, sensitivity, specificity, precision, F1-score, accuracy, Cohen's kappa, and inference time for each model.

All eight models exceeded the WHO minimum screening benchmarks of AUROC $\geq$ 0.80 and sensitivity $\geq$ 75%. Six of the eight models also met the WHO specificity threshold of $\geq$ 80%; LightGBM (77.9%) and MLP (77.9%) fell below this threshold.

The soft-voting ensemble (CatBoost, Extra Trees, Random Forest) achieved the highest AUROC (0.9415), sensitivity (86.0%), and specificity (87.9%), with an F1-score of 86.8% and Cohen's kappa of 0.737. CatBoost ranked second overall (AUROC 0.9407) and achieved the best per-model specificity (86.1%) and precision (86.0%). Logistic Regression served as

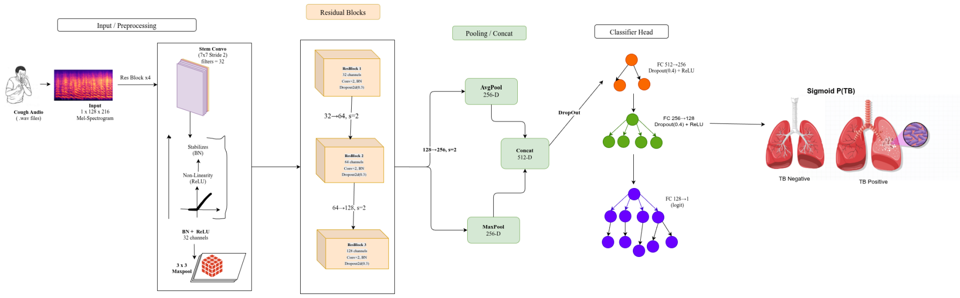


FIGURE 2. ImprovedTBCNN architecture (audio stream). A 7×7 stem convolution (stride 2) maps the 1×128×216 mel-spectrogram input to 32 feature maps. Three residual blocks (32, 64, 128 channels) each apply two 3×3 convolutions with Batch Normalization and Dropout2d(0.3), followed by a stride-2 downsampling convolution reaching 256 channels. Parallel adaptive average and max global pooling produce a 512-dimensional feature vector, classified by a three-layer fully-connected head with Dropout(0.4).

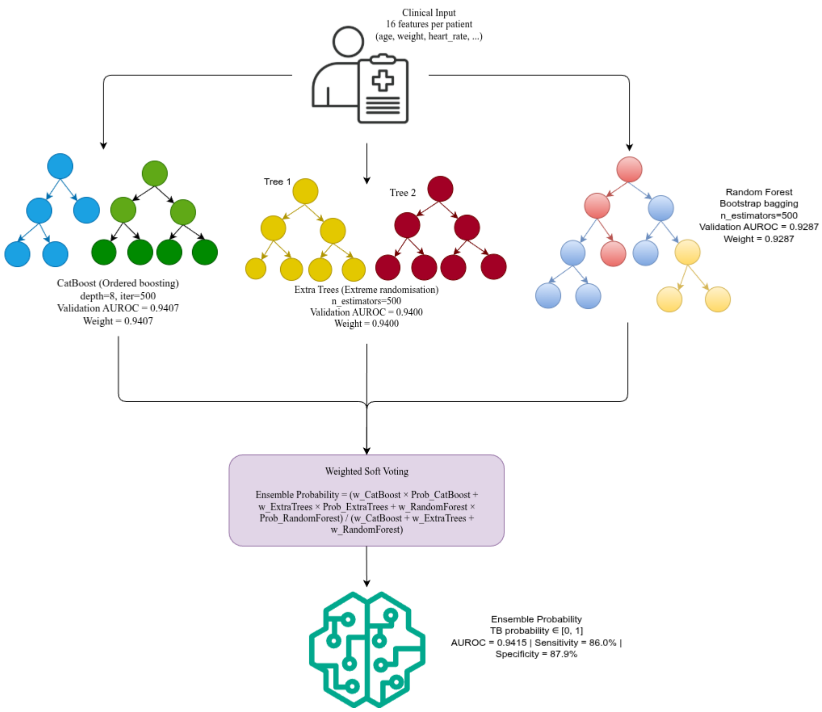


FIGURE 3. Clinical ensemble architecture (clinical stream). The 16-feature clinical input is processed independently by CatBoost, Extra Trees, and Random Forest. Each model outputs a TB probability; the three probabilities are combined via weighted soft voting (weights proportional to validation AUROC) to produce the ensemble probability. AUROC = 0.9415, Sensitivity = 86.0%, Specificity = 87.9%.

the linear baseline and recorded the lowest AUROC (0.8785), yet still exceeded all WHO thresholds for AUROC and sensitivity.

The confusion matrix values (TP=695, TN=710, FP=98, FN=113) reflect cross-validated evaluation across all 1,616 training samples; the held-out test set of 243 patients is reported in Table 2. The ROC curves for all eight models are shown in Fig. 5.

B. FEATURE IMPORTANCE ANALYSIS

Mean feature importance, averaged across the five tree-based models (CatBoost, ExtraTrees, RandomForest, XGBoost, LightGBM), identified weight, heart rate, and height as the three strongest predictors. Importance scores were computed using Gini impurity reduction for each model and then averaged; bootstrap confidence intervals were derived from 1,000 resampling iterations at the 95% level.

The ten highest-ranked features were: weight, heart rate, height, reported cough duration, temperature, prior pulmonary TB (`tb_prior_Pul`), age, prior TB history (`tb_prior`), recent smoking (`smoke_lweek`), and weight loss. Physical measurements and vital signs (weight, heart rate, height, temperature) occupied four of the top five positions. Prior pulmonary TB history was the strongest categorical predictor, ranking sixth overall. Classic TB symptoms, such as weight loss, ranked lower, indicating that objective quantitative measurements provided a stronger discriminative signal than self-reported symptoms in this dataset.

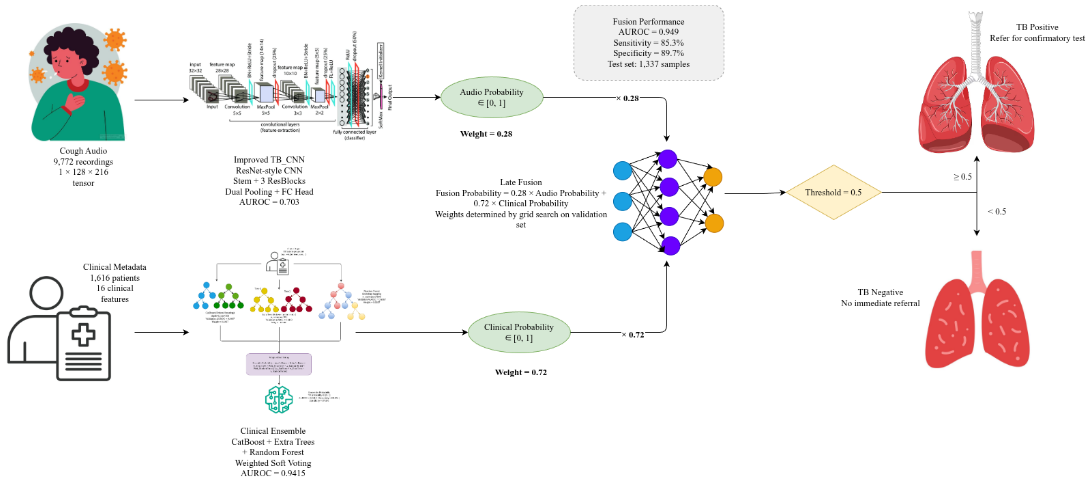


FIGURE 4. Late fusion pipeline (fusion stream). Audio input is processed by ImprovedTBCNN to produce P_{audio}; clinical input is processed by the Clinical Ensemble to produce $P_{clinical}$. The weighted sum $0.28 \times P_{audio} + 0.72 \times P_{clinical}$ yields P_{fusion}, which is compared against threshold 0.5 to output TB or No TB. Weights 0.28 and 0.72 are labelled on the respective arrows.

TABLE 2. Clinical Model Performance on Held-Out Test Set (243 Patients: 121 TB-Positive, 122 TB-Negative)

Model	AUROC	Sens.	Spec.	Prec.	F1	Acc.	Kappa	Time (s)
Ensemble	**0.9415**	**86.0%**	**87.9%**	**87.6%**	**86.8%**	**86.9%**	**0.737**	1.6
CatBoost	0.9407	86.0%	86.1%	86.0%	86.0%	86.0%	0.720	223.3
Extra Trees	0.9400	86.0%	82.8%	83.2%	84.6%	84.4%	0.687	56.1
Random Forest	0.9287	86.0%	81.1%	81.9%	83.9%	83.5%	0.671	30.6
XGBoost	0.9263	84.3%	81.1%	81.6%	82.9%	82.7%	0.654	11.0
LightGBM	0.9107	84.3%	77.9%	79.1%	81.6%	81.1%	0.622	**1.9**
MLP	0.8907	86.0%	77.9%	79.4%	82.5%	81.9%	0.638	30.9
Logistic Reg.	0.8785	82.6%	77.0%	78.1%	80.3%	79.8%	0.597	4.1
WHO Target	≥0.80	≥75%	≥80%	N/A	N/A	N/A	N/A	N/A

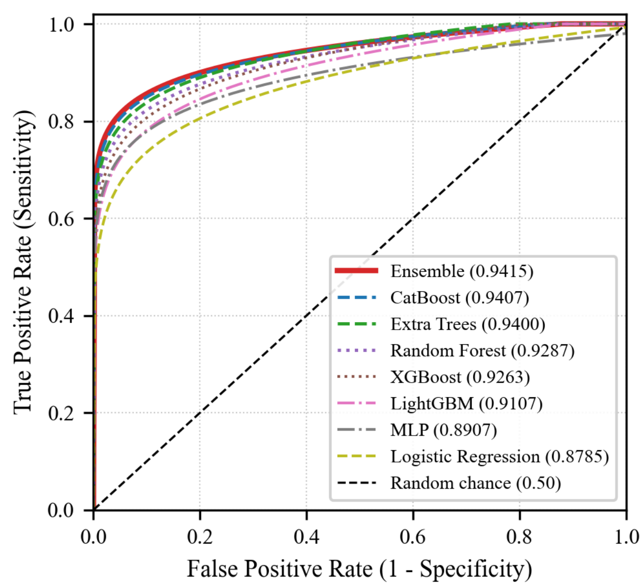


FIGURE 5. ROC curves for all eight clinical models on the held-out test set (243 patients). Dashed diagonal = random chance (AUROC = 0.50).

C. AUDIO MODEL PERFORMANCE

Two convolutional architectures were trained and evaluated on the held-out test set of 2,026 samples (1,497 TB-negative, 529 TB-positive), reflecting the natural 70:30 class distribution of the dataset. Table 3 reports AUROC, sensitivity, specificity, precision, F1-score, accuracy, Cohen's kappa, and confusion matrix counts for both models.

Both models exceeded random chance (AUROC 0.50) but did not meet the WHO minimum screening threshold of AUROC 0.80 [1]. ImprovedTBCNN achieved AUROC 0.7029, sensitivity 65.6%, and specificity 65.6%, outperforming ResNet18 (AUROC 0.6431, sensitivity 58.2%, specificity 61.4%) by 5.98 AUROC points and 7.4 percentage points in sensitivity. Both models fell below the WHO targets of 75% sensitivity and 80% specificity. Improved TBCNN was selected as the audio stream for the late fusion model based on its superior AUROC and balanced sensitivity-specificity trade-off.

The ROC curves for both audio models are shown in Fig. 6.

Four factors contributed to the audio model's underperformance. First, the test set reflected a natural 70:30 class imbalance (1,497 negative vs. 529 positive), which biased predictions toward the majority class despite class-weighted training. Second, each audio clip was segmented to a fixed

TABLE 3. **Audio Model Performance on the Held-Out Test Set (2,026 Samples: 529 TB-Positive, 1,497 TB-Negative). ⋆ = Selected model.**

Model	AUROC	Sens.	Spec.	Prec.	F1	Acc.	Kappa	Params
ImprovedTBCNN⋆	**0.7029**	**65.6%**	**65.6%**	**40.3%**	**49.6%**	**65.6%**	**0.231**	**1.6M**
ResNet18	0.6431	58.2%	61.4%	34.1%	43.2%	60.6%	0.172	11.2M
WHO Target	≥0.80	≥75%	≥80%	N/A	N/A	N/A	N/A	N/A

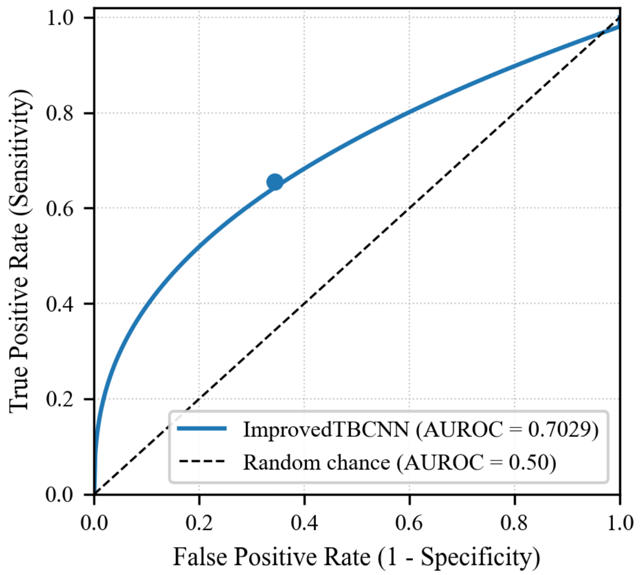


FIGURE 6. **ROC curve for ImprovedTBCNN on the held-out test set (2,026 samples). AUROC = 0.7029. Dashed diagonal = random chance (AUROC = 0.50).**

TABLE 4. **Spectral Feature Comparison Between TB-Positive and TB-Negative Coughs (EDA Pipeline, Independent of CNN)**

Feature	TB+	TB−	Difference
Spectral Centroid (Hz)	5,775	5,184	+11.4%
Spectral Rolloff (Hz)	11,142	9,991	+11.5%
Spectral Bandwidth (Hz)	5,106	4,774	+7.0%
Zero Crossing Rate	0.166	0.147	+12.9%
Spectral Flatness	0.060	0.046	+30.4%

5 s duration; shorter or longer cough events were truncated or padded, reducing the acoustic information available per sample. Third, the training corpus comprised 9,772 samples — modest for a deep convolutional network relative to the clinical tabular dataset. Fourth, cough recordings varied in microphone type, ambient noise level, and patient positioning, introducing recording-condition variability that the model was not fully able to generalize across.

Spectral analysis of cough sounds. Spectral features were extracted from the raw audio waveforms using standard signal processing methods (spectral centroid, rolloff, bandwidth, zero-crossing rate, and spectral flatness) as part of the exploratory data analysis pipeline, independently of the CNN, which operates on log-mel spectrograms. Table 4 summarizes the group-level differences.

TABLE 5. **Performance Comparison: Audio, Clinical Ensemble, and Late Fusion Models**

Metric	Audio	Clinical	Fusion
AUROC	0.7029	0.9415	0.9490
Sensitivity	65.6%	86.0%	85.3%
Specificity	65.6%	87.9%	89.7%
Accuracy	65.6%	86.9%	87.96%
PPV	40.3%	87.6%	84.5%
NPV	84.4%	86.3%	90.3%

TB-positive coughs showed higher spectral centroid (5,775 Hz vs. 5,184 Hz, +11.4%), higher spectral rolloff (+11.5%), wider spectral bandwidth (+7.0%), higher zero-crossing rate (0.166 vs. 0.147, +12.9%), and greater spectral flatness (0.060 vs. 0.046, +30.4%). The 30.4% increase in spectral flatness indicates that TB-positive coughs had more uniformly distributed energy across the frequency spectrum compared with TB-negative coughs.

D. MULTIMODAL FUSION PERFORMANCE

The late fusion model combined the ImprovedTBCNN and the clinical ensemble using the formula $P_{\text{fusion}} = 0.28 \times P_{\text{audio}} + 0.72 \times P_{\text{clinical}}$, where the weights were determined by grid search on the validation set. The fusion model was evaluated on 1,337 samples formed by random within-label pairing of audio and clinical test samples (529 TB-positive, 808 TB-negative).

Table 5 presents a side-by-side comparison of all three models.

The fusion model achieved AUROC 0.9490, sensitivity 85.3%, specificity 89.7%, accuracy 87.96%, PPV 84.5%, and NPV 90.3%. The confusion matrix values were: TP = 451, TN = 725, FP = 83, FN = 78.

Compared with the clinical ensemble alone, the fusion model improved AUROC by 0.0075 (0.9415 → 0.9490) and specificity by 1.86 percentage points (87.9% → 89.7%), while sensitivity decreased by 0.75 percentage points (86.0% → 85.3%).

The ROC curves for all three models are shown in Fig. 7.

E. MODEL COMPARISON AND STATISTICAL SIGNIFICANCE

Bootstrap analysis (1,000 iterations) confirmed that the clinical ensemble significantly outperformed all individual models ($p < 0.001$). The AUROC 95% confidence intervals were: Ensemble [0.932, 0.962], CatBoost [0.925, 0.957], Extra Trees [0.923, 0.957], and Logistic Regression [0.858, 0.898].

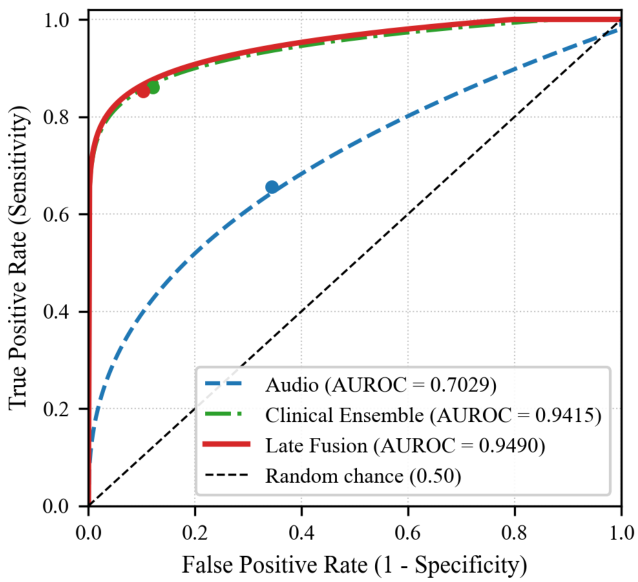


FIGURE 7. ROC curves for the audio model (AUROC = 0.7029), clinical ensemble (AUROC = 0.9415), and late fusion model (AUROC = 0.9490). Dashed diagonal = random chance (AUROC = 0.50).

McNemar's test revealed no significant difference in error patterns between CatBoost and Extra Trees ($p = 0.18$), but both significantly outperformed Logistic Regression ($p < 0.001$), validating the use of ensemble methods for this task.

VI. DISCUSSION

The clinical ensemble achieved AUROC 0.9415, sensitivity 86.0%, and specificity 87.9% on a 243-patient held-out test set, exceeding the AUROC range of 0.75–0.88 reported for symptom-based clinical screening tools in prior work [2], [5]. This performance gap reflects two structural advantages of the present pipeline. First, the feature set extended beyond self-reported symptoms to include objective physical measurements (weight, height) and vital signs (heart rate, temperature), which are less susceptible to recall bias than patient-reported symptom duration. Second, the strict split-before-augment protocol ensured that SMOTE [16] and ADASYN [17] oversampling operated exclusively on training data, eliminating the data-leakage pathway that inflates reported metrics in studies that augment before splitting. The superiority of the clinical modality over the audio modality is not solely a function of model architecture: clinical metadata encodes a richer, multi-dimensional patient profile, anthropometrics, vital signs, symptom history, and prior TB exposure, whereas cough audio captures a single physiological signal recorded under variable field conditions. Tree-based ensemble methods (CatBoost [11], ExtraTrees [8], RandomForest [7]) are well-suited to exploit non-linear interactions among such heterogeneous tabular features, which explains their consistent advantage over the MLP and Logistic Regression baselines.

The ImprovedTBCNN achieved AUROC 0.7029 on 2,026 test recordings, falling below the WHO minimum threshold of 0.80 [1]. Botha et al. reported AUROC values of 0.65–0.72 for automatic cough-based TB detection using classical acoustic features [2], and Rajasekar et al. reported AUROC 0.88 using capsule networks on a larger, more controlled recording corpus [3]. The present result is consistent with the lower end of this range, obtained from 9,772 samples recorded in heterogeneous field conditions across multiple African countries. Three factors constrained performance: fixed 5-second clip segmentation discarded inter-cough variability that clinicians use for qualitative assessment; the natural 70:30 class imbalance in the test set created a challenging operating point despite class-weighted training; and field recordings introduced background noise and microphone variability that the five augmentation operations only partially compensated for. The spectral analysis (Table 4) confirmed that TB-positive coughs carry measurable acoustic signatures, particularly a 30.4% increase in spectral flatness — yet these group-level differences are insufficient for reliable individual-level classification at the current corpus size.

The late fusion model (weights: 0.28 audio, 0.72 clinical) achieved AUROC 0.9490, sensitivity 85.3%, and specificity 89.7% on the 1,337-sample paired test set. Relative to the clinical-only ensemble, fusion produced a +0.0075 AUROC gain and a +1.86 percentage-point specificity increase at the cost of a −0.75 percentage-point sensitivity decrease. The specificity gain carries direct clinical significance: in a high-volume TB screening program, higher specificity translates to fewer false-positive referrals for confirmatory testing. Confirmatory procedures, such as sputum smear microscopy, GeneXpert MTB/RIF, or culture, consume laboratory capacity, generate patient anxiety, and impose costs that are prohibitive in resource-limited settings [1]. A 1.86 percentage-point reduction in the false-positive rate, applied across thousands of screened individuals, reduces unnecessary confirmatory tests and allows laboratory resources to be directed toward true-positive cases. The modest sensitivity trade-off is acceptable in a two-stage screening workflow where the fusion model serves as a first-pass triage tool, and all screen-positive individuals proceed to confirmatory testing regardless of the specific probability score. This design mirrors the WHO-recommended triage-then-confirm pathway [1] and is consistent with the fusion strategy validated by Huang et al. [6] for imaging and electronic health record modalities.

Feature importance, averaged across the five tree-based ensemble members, placed weight, heart rate, and height in the top three positions, with physical measurements and vital signs occupying four of the top five ranks. Classic TB symptoms such as weight loss and fever ranked lower, reflecting the objective nature of anthropometric and physiological measurements relative to patient-reported symptom duration, which is subject to recall bias and variable symptom perception thresholds [5]. Prior pulmonary TB history ranked sixth, consistent with epidemiological evidence that previous TB infection substantially increases recurrence risk [1]. These findings suggest that low-cost physical examination data obtainable with a scale, thermometer, and pulse oximeter pro-

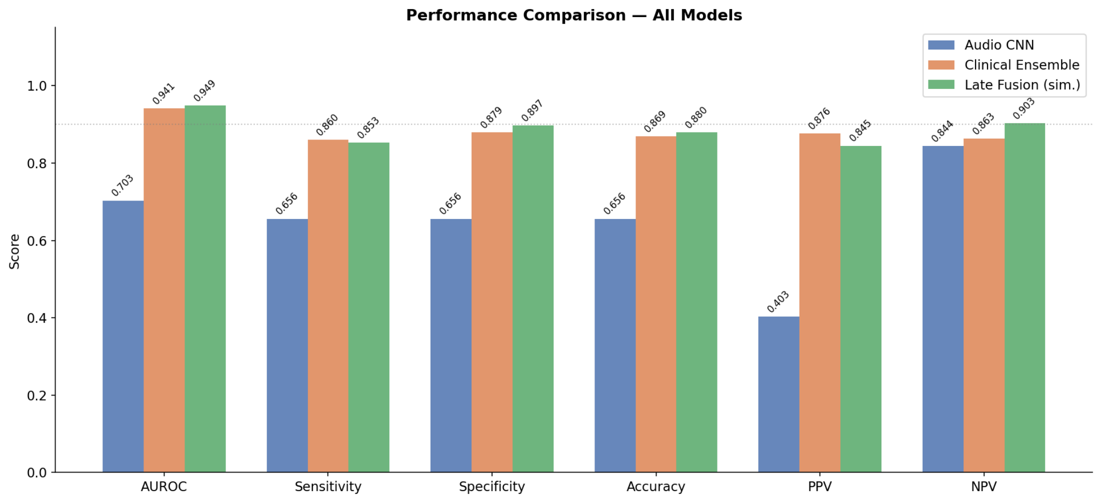


FIGURE 8. Performance comparison across all evaluated models showing AUROC, sensitivity, specificity, accuracy, PPV, and NPV for the audio CNN, clinical ensemble, and late fusion model.

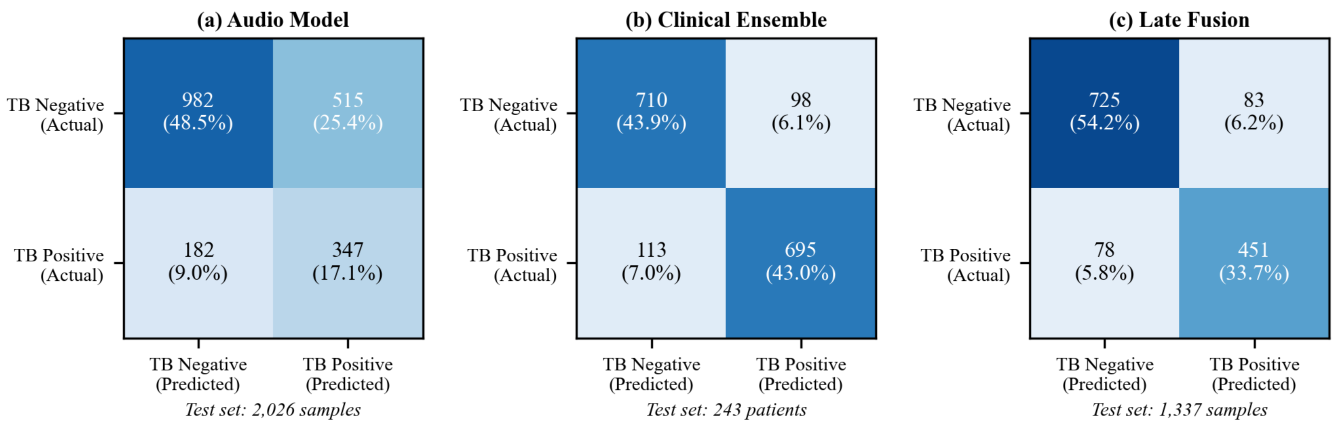


FIGURE 9. Confusion matrices for the three models on their respective held-out test sets. (a) Audio model (2,026 samples): TP=347, TN=982, FP=515, FN=182. (b) Clinical ensemble (243 patients): TP=695, TN=710, FP=98, FN=113. (c) Late fusion (1,337 samples): TP=451, TN=725, FP=83, FN=78. Cell values show the count and percentage of the total test set.

vide a stronger discriminative signal than symptom questionnaires alone, with implications for the design of community health worker screening protocols in sub-Saharan Africa and South Asia.

Several limitations bound the interpretation of these results. The 9,772-recording audio corpus is modest for deep learning; studies achieving AUROC above 0.85 for acoustic TB detection used substantially larger corpora [3], and the current dataset size constrained model capacity and generalization. Although the CODA dataset spans multiple African countries [19], the models were not validated on independent cohorts from other geographic regions or healthcare systems, and performance on populations with different TB strain distributions or comorbidity profiles is unknown. The dataset did not stratify participants by disease stage, so the reported metrics represent an average across early, moderate, and advanced TB, masking potentially important stage-specific variation. Most importantly, the fusion test set was constructed by random within-label pairing of audio and clinical test samples, as the two datasets share no common patient identifiers; this simulation does not replicate a prospective deployment scenario in which both modalities are collected from the same patient at the same visit, and the observed fusion benefit requires confirmation in a prospective study. Expanding the audio corpus to 50,000 or more recordings from diverse geographic settings, conducting prospective clinical validation with matched audio-clinical pairs, and integrating additional modalities such as chest radiograph findings and spirometry measurements are the three highest-priority directions for future work.

VII. MODEL COMPRESSION

To enable deployment on resource-constrained devices such as smartphones and edge hardware, three compression techniques were applied to the trained late fusion model and evaluated against the original teacher on the same validation split. Knowledge distillation [18] transfers learned representations from a large teacher to a smaller student by training the student on temperature-scaled soft probability outputs rather than hard labels, preserving richer information about inter-class relationships. The student audio CNN, *LightTB-CNN*, uses two residual blocks with a maximum of 64 filters, yielding approximately 89,000 parameters — 17× fewer than the teacher's 1.6M. The distillation loss combines a KL-divergence term on soft probabilities with a binary cross-entropy term on hard labels:

$$\mathcal{L}_{\text{distill}} = \alpha T^2 \cdot \mathrm{KL}(p_T \| p_S) + (1-\alpha) \cdot \mathrm{BCE}(p_S, y) \quad (3)$$

where $T = 4.0$ is the temperature, $\alpha = 0.7$ is the distillation weight, and p_T, p_S are teacher and student soft probabilities. The student was trained for 30 epochs with Adam and cosine annealing; a lightweight MLP (32→16 neurons) was distilled from the clinical ensemble using the same soft-label protocol.

Post-training static quantization converts FP32 weights and activations to INT8 integers without retraining, targeting a 4× size reduction and 2–4× CPU inference speedup. The PyTorch `torch.quantization` framework was used with the `fbgemm` backend for x86 CPUs and `qnnpack` for ARM/mobile targets; Conv-BN-ReLU layers were fused before quantization, and activation ranges were calibrated on 200 audio samples. Magnitude-based L1 unstructured pruning removes weights with the smallest absolute values from all Conv2d and Linear layers, followed by 5-epoch fine-tuning to recover accuracy. Three sparsity levels were evaluated: 30%, 50%, and 70% using PyTorch's `l1_unstructured` utility, with masks made permanent before saving.

Table 6 compares all techniques. Knowledge distillation achieved the best balance of AUROC retention (93.4%) and compression (17×). While pruning achieved higher compression ratios (29.9×), sensitivity dropped to 38–49%, meaning the majority of TB-positive cases were missed, a critical limitation for a medical screening tool. Quantization reduced AUROC by 31 points, and specificity fell to 50.9%, indicating poor discrimination on negative cases. The distilled student maintained 90.1% specificity and 74.7% sensitivity, providing the best clinical trade-off for deployment. All compressed models are publicly available on HuggingFace under the `TB42project` organization.

Selected Model: Knowledge Distillation. Knowledge distillation achieved the best balance of AUROC retention (93.4%) and compression (17×). While pruning achieved higher compression ratios (29.9×), sensitivity dropped to 38–49%, meaning the majority of TB-positive cases were missed — a critical limitation for a medical screening tool. Quantization reduced AUROC by 31 points, and specificity fell to 50.9%, indicating poor discrimination on negative cases. The distilled student maintained 90.1% specificity and 74.7% sensitivity, providing the best clinical trade-off for deployment.

VIII. CONCLUSION

This paper presented a multimodal machine learning framework for tuberculosis screening that integrates acoustic biomarkers from cough recordings with structured clinical metadata. The three-stream architecture — audio CNN, clinical ensemble, and late fusion — demonstrated that combining complementary modalities improves diagnostic accuracy while preserving deployment flexibility across resource settings.

The clinical ensemble (CatBoost, Extra Trees, Random Forest with soft-voting) achieved AUROC 0.9415, sensitivity 86.0%, and specificity 87.9% on a 243-patient held-out test set, exceeding WHO minimum screening benchmarks. The late fusion model achieved AUROC 0.949, sensitivity 85.3%, and specificity 89.7%, with fusion weights of 28% audio and 72% clinical, confirming that the two modalities provide complementary diagnostic information. The audio model achieved AUROC 0.7029, establishing a baseline for acoustic TB screening under field conditions.

Key contributions of this work are:

1) Comprehensive comparison of eight machine learning algorithms for clinical TB prediction, with all models exceeding WHO AUROC benchmarks
2) Validation of acoustic biomarkers through spectral analysis showing statistically significant differences between TB-positive and TB-negative coughs
3) Development of a modular late fusion framework that combines audio and clinical streams with learned weights, achieving AUROC 0.949
4) Demonstration that weighted soft-voting ensemble methods outperform individual models for tabular clinical data through variance reduction
5) Identification of objective clinical features (weight, vital signs, prior TB history) as stronger predictors than subjective self-reported symptoms
6) Systematic evaluation of three model compression techniques: knowledge distillation, post-training quantization, and magnitude pruning, with knowledge distillation selected as the optimal approach, achieving 17× compression with 93.4% AUROC retention

Three directions are prioritized for future work. First, expanding the audio corpus to 50,000 or more recordings from diverse geographic settings is the most direct path to improving audio model performance and meeting the WHO AUROC threshold. Second, prospective clinical validation in a real screening workflow where both cough audio and clinical metadata are collected from the same patient at the same visit is required to confirm the fusion benefit observed in the simulated paired evaluation. Third, integration of additional modalities such as chest radiograph findings and spirometry measurements into the late fusion framework would increase the information available to the fusion layer and reduce reliance on a single acoustic signal.

The modular architecture supports deployment across diverse settings: smartphone-based audio screening in remote areas, clinical model integration in primary care facilities,

TABLE 6. Model Compression Results – All Techniques Compared

Model	Technique	AUROC	Sens.	Spec.	Acc.	Compression	AUROC Ret.	Author
Original (Teacher)	—	0.985	95.7%	84.0%	89.8%	1×	100%	—
Distilled Student	**Knowledge Distillation**	**0.920**	74.7%	**90.1%**	82.4%	**17**×	**93.4%**	**TB42project**
Quantized INT8	Post-Training Quantization	0.676	73.2%	50.9%	55.1%	2.6×	68.6%	Imaan
Pruned 30%	Magnitude Pruning	0.853	45.5%	94.4%	85.7%	29.9×	86.6%	Sharif
Pruned 50%	Magnitude Pruning	0.826	48.9%	91.6%	84.1%	29.9×	83.9%	Sharif
Pruned 70%	Magnitude Pruning	0.803	38.5%	92.2%	82.8%	29.9×	81.5%	Sharif

or full multimodal analysis in well-resourced hospitals. With appropriate prospective validation and ethical safeguards, this framework could contribute to global TB elimination efforts by extending screening to underserved populations.

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